



Tertiary Infotech Academy Pte Ltd

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LAB 10 INSTRUCTIONS

For

Verify link-local resolution

Course Code: C1123

Conducted by

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Version 5.1

Lab 10 — Verify link-local resolution

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 10 — ARP Traffic Analysis

Learning outcome: LO3 — Interpret DNS, ARP, IPv4, ICMP and UDP evidence.

Time: about 50 minutes

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Scenario

A technician suspects an ARP problem on the branch LAN. You check what the capture actually proves before anyone raises a spoofing alert.

Goal

ARP requests are link-local broadcasts; replies usually return to the requester.

What you will produce

A verified ARP table entry and an evidence-based conclusion.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to <code>outputs/evidence.csv</code>
outputs/findings.md	Findings sheet you complete
assets/arp-table.csv	ARP table you complete
assets/arp-hypotheses.md	Competing explanations to test

Lab at a glance

1. Record ARP request and reply fields
2. Confirm the server IP-to-MAC mapping
3. Verify the opcode in the packet bytes
4. Test the hypotheses against the evidence

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap`. Apply `arp`. Expand the request and reply, then record sender IP, sender MAC, target IP and opcode.

3. Apply `arp.opcode == 2`. Confirm the advertised server mapping is 192.0.2.20 to 02:00:00:00:00:20.

4. Use the packet bytes to verify the opcode field is 2 in the reply. Add the mapping to `assets/arp-table.csv`.

5. Compare `assets/arp-hypotheses.md` with the evidence. State that this capture shows one successful exchange, without inventing repeated failures or conflicting MACs.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

arp.opcode == 2					
No.	Time	Source	Destination	Protocol	Info
2	0.010000	02:00:00:00:00:20	02:00:00:00:00:10	ARP	192.0.2.20 is at 02:00:00:00:00:20
branch-office.pcap Displayed: 1					

Lab 10 expected evidence

Test it

The supplied `arp.opcode == 2` expression matches 1 frame(s) in the specified capture. Run `scripts/verify.py` and compare the listed frame numbers. Keep your findings and exported table in `outputs/`. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching `lab-tls.keys` file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -Y arp -T fields -e arp.opcode -e arp.src.proto_ipv4 -e arp.src.hw_mac
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

Ethernet and ARP lab — read the ARP cache on your own machine (arp -a) and match it to captured replies. Source: J.F. Kurose and K.W. Ross, Wireshark Labs (gaia.cs.umass.edu/kurose_ross/wireshark.php).

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.